Machine Learning

Anomaly Detection

Finding unusual events

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Outline

- Anomaly Detection Example
- Gaussian (Normal) Distribution
- Algorithm
- Developing and evaluating an anomaly detection system
- Anomaly detection vs supervised learning
- Choosing what features to use

Anomaly detection example

Aircraft engine features:

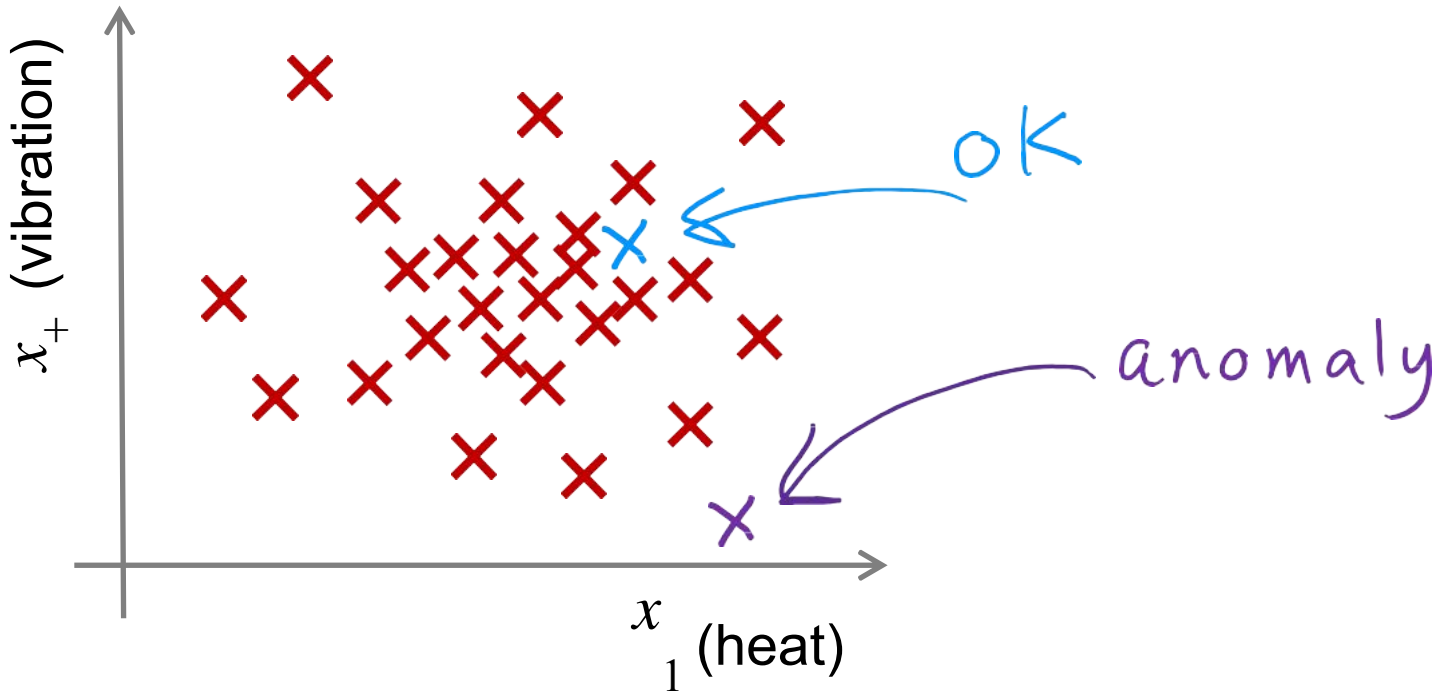
x_1 = heat generated

x_2 = vibration intensity

…

Dataset: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$

New engine: x_{test}



Density estimation

Dataset: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$

probability of x being seen in dataset

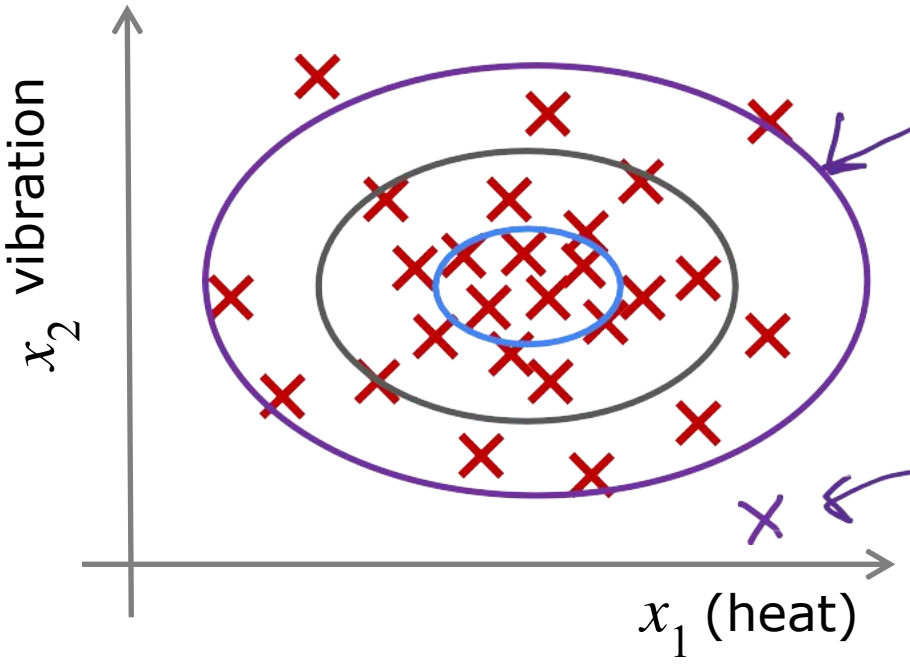
Model $p(x)$

Is x_{test} anomalous?

high probability

$p(x_{test}) \geq \varepsilon$

ok (normal)



low probability

$p(x_{test}) < \varepsilon$

epsilon: small number

unlikely

anomaly

Anomaly detection example

how often log in?
how many web pages visited?
transactions?
posts? typing speed?

Fraud detection:

- $x^{(*)}$ = features of user i's activities
- Model $p(x)$ from data.
- Identify unusual users by checking which have $p(x) < \varepsilon$

perform additional checks to identify real fraud vs. false alarms

Manufacturing:

$x^{(*)}$= features of product i

airplane engine
circuit board
smartphone

Monitoring computers in a data center:

$x^{(i)}$= features of machine i

- x_1 = memory use,
- x_2= number of disk accesses/sec,
- x_3 = CPU load,
- x_4= CPU load/network traffic.

ratios



Gaussian (Normal) Distribution

Gaussian (Normal) distribution

σ standard deviation

σ^2 variance

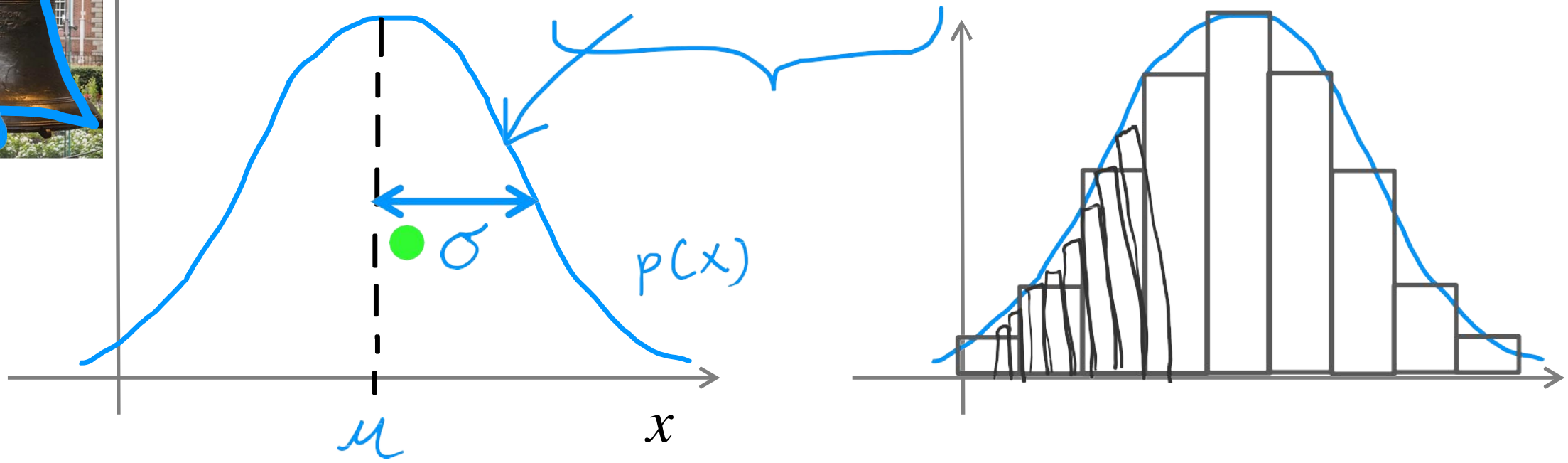
Say x is a number.

Probability of x is determined by a Gaussian with mean μ, variance σ^2.

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

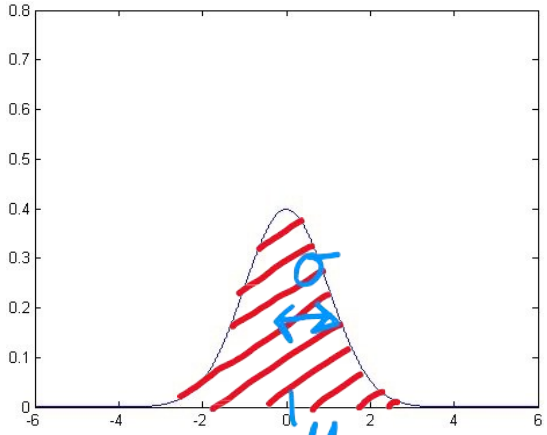
$\pi = 3.14$



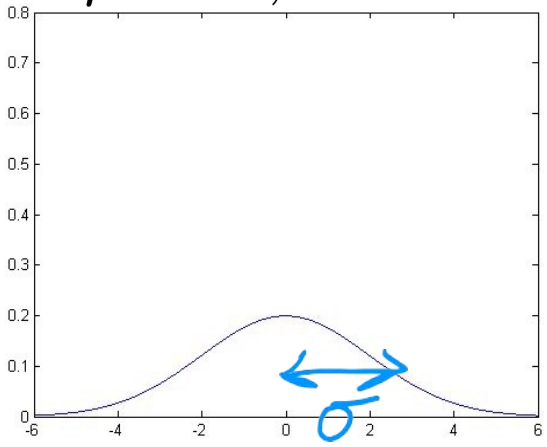


Gaussian distribution example

$\mu = 0, \sigma = 1$

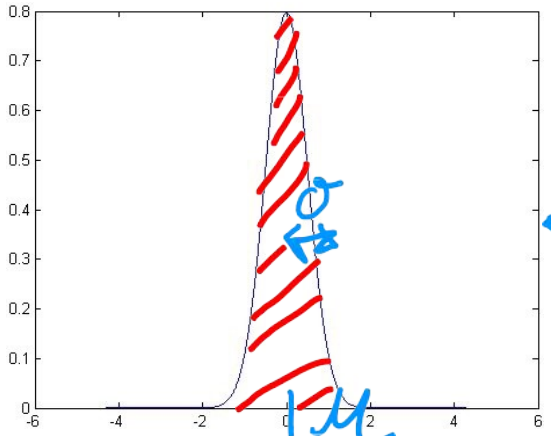


$\mu = 0, \sigma = 2$



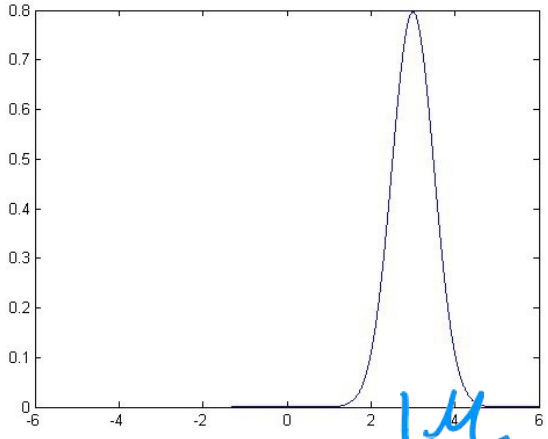
$\sigma^2 = 4$

$\mu = 0, \sigma = 0.5$



$\sigma^2 = 0.25$

$\mu = 3, \sigma = 0.5$

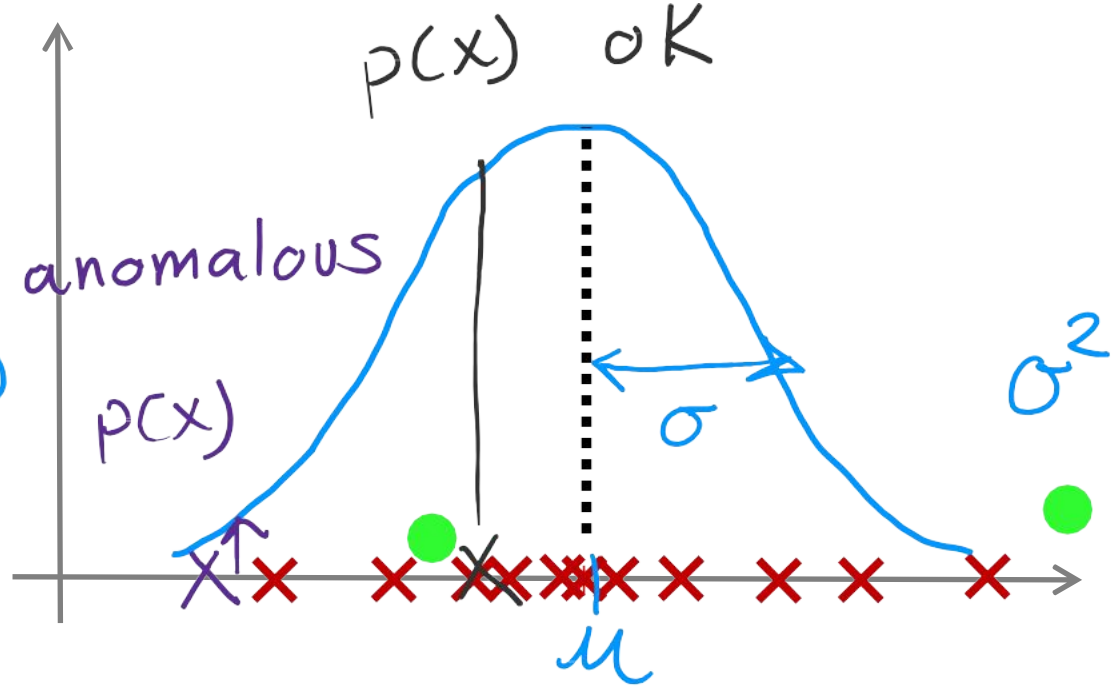


Parameter estimation

Dataset: $\{x^{(1)}, x^{(+)}, \ldots, x^{(m)}\}$

maximum likelihood for μ, σ

$\frac{1}{m-1}$



$$\mu = \frac{1}{m}\sum_{i=1}^{m} x^{(i)}$$

$$\sigma^2 = \frac{1}{m}\sum_{i=1}^{m} (x^{(i)} - \mu)^2$$

x is a number (1 feature)

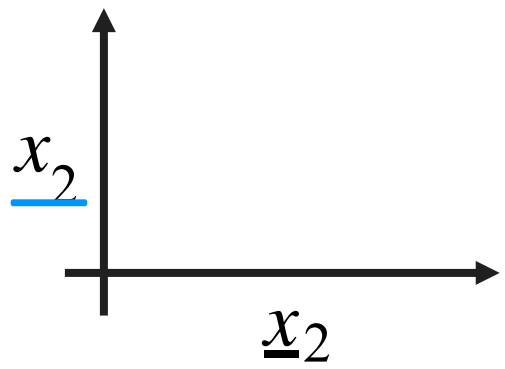


Algorithm

Density estimation

Training set: $\{\vec{x}^{(1)}, \vec{x}^{(2)}, \ldots, \vec{x}^{(m)}\}$

Each example $\vec{x}^{(i)}$ has n features



$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix}$$

$$p(\vec{x}) = p(x_1; \mu_1, \sigma_1^2) * p(x_2; \mu_2, \sigma_2^2) * p(x_3; \mu_3, \sigma_3^2) * \cdots * p(x_n; \mu_n, \sigma_n^2)$$

$$= \prod_{j=1}^{n} p(x_j; \mu_j, \sigma_j^2)$$

$\sum$ "add"

$$p(x_1 = \text{high temp}) = 1/10$$

$$p(x_2 = \text{high vibra}) = 1/20$$

$$p(x_1, x_2) = p(x_1) * p(x_2)$$

$$= \frac{1}{10} \times \frac{1}{20} = \frac{1}{200}$$

Anomaly detection algorithm

1. Choose n features x_i that you think might be indicative of anomalous examples.

2. Fit parameters $\mu_1, \ldots, \mu_n, \sigma_1^2, \ldots, \sigma_n^2$

$$\mu_j = \frac{1}{m}\sum_{i=1}^{m} x_j^{(i)} \qquad \sigma_j^2 = \frac{1}{m}\sum_{i=1}^{m}(x_j^{(i)} - \mu_j)^2$$

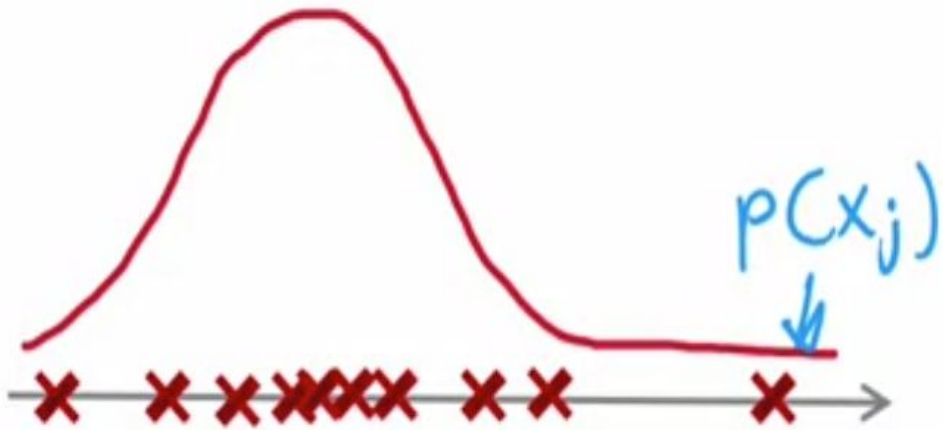
Vectorized formula

$$\vec{\mu} = \frac{1}{m}\sum_{i=1}^{m} \vec{x}^{(i)} \qquad \vec{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \cdots \\ \mu_n \end{bmatrix}$$

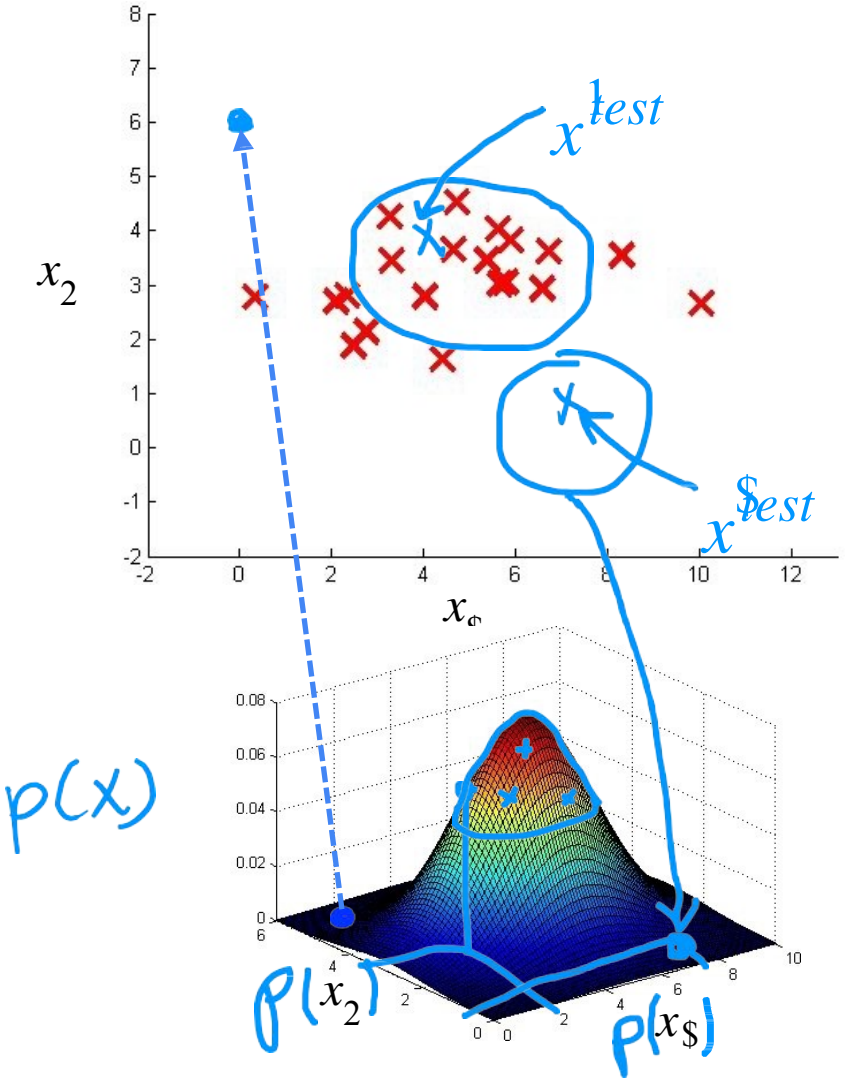
3. Given new example x, compute $p(x)$:

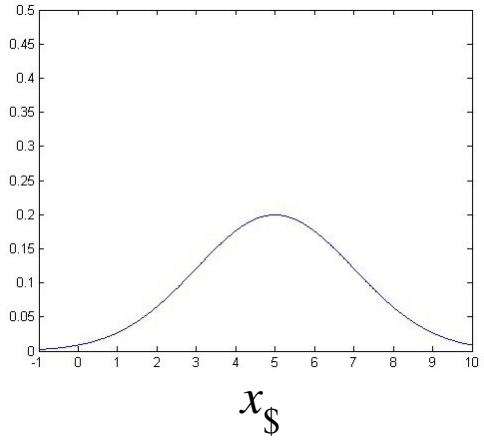
$$p(x) = \prod_{j=1}^{n} p(x_j; \mu_j, \sigma_j^2) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}\sigma_j} exp(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2})$$

Anomaly if $p(x) < \varepsilon$



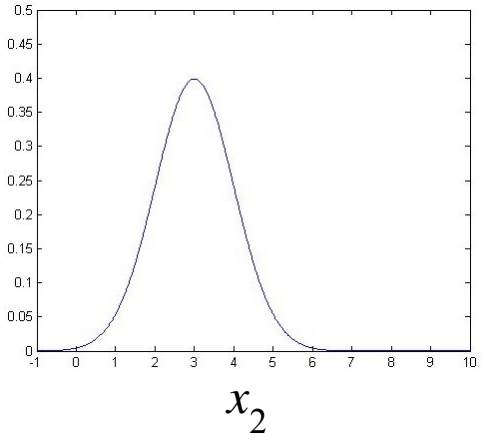
Anomaly detection example





$$\mu_1 = 5, \sigma_1 = 2$$

$$p(x_1; \mu_1, \sigma_1^2)$$



$$\mu_2 = 3, \quad \sigma_2 = 1$$

$$p(x_2; \mu_2, \sigma_2^2)$$

$$\varepsilon = 0.02$$

$$p\left(x^{(1)}_{test}\right) = 0.0426 \longrightarrow \text{"ok"}$$

$$p\left(x^{(2)}_{test}\right) = 0.0021 \longrightarrow \text{anomaly}$$



Developing and evaluating an anomaly detection system

The importance of real-number evaluation

When developing a learning algorithm (choosing features, etc.),
making decisions is much easier if we have a way of evaluating our learning algorithm.

Assume we have some labeled data, of anomalous and non-anomalous examples.

$y = 1$ $y = 0$

Training set: $x^{(1)}, x^{(2)}, \dots, x^{(m)}$ (assume normal examples/not anomalous)

$y = 0$ for all training examples

Cross validation set: $\left(x_{cv}^{(1)}, y_{cv}^{(1)}\right), \dots, \left(x_{cv}^{(m_{cv})}, y_{cv}^{(m_{cv})}\right)$

Test set: $\left(x_{test}^{(1)}, y_{test}^{(1)}\right), \dots, \left(x_{test}^{(m_{test})}, y_{test}^{(m_{test})}\right)$

include a few anomalous examples $y = 1$

mostly normal examples $y = 0$

Aircraft engines monitoring example

10000 good (normal) engines
20 flawed engines (anomalous)

2 to 50

$y=0$

$y=1$

Training set: 6000 good engines
CV: 2000 good engines ($y=0$)

train algorithm on training set

10 anomalous ($y=1$)

use cross validation set

tune ε tune x_j

Test: 2000 good engines ($y=0$),

10 anomalous ($y=1$)

Alternative: No test set

Use if very few labeled anomalous examples

higher risk of overfitting

Training set: 6000 good engines

CV: 4000 good engines ($y=0$), ~~20~~ 2 anomalous ($y=1$)

tune ε tune x_j

Algorithm evaluation

skewed datasets

Fit model $p(x)$ on training set $x^{(1)}, x^{(2)}, \ldots, x^{(m)}$
On a cross validation/test example x, predict

$$y = \begin{cases} 1 & \text{if } p(x) < \varepsilon \text{ (anomaly)} \\ 0 & \text{if } p(x) \geq \varepsilon \text{ (normal)} \end{cases}$$

10

2000

Possible evaluation metrics:

- True positive, false positive, false negative, true negative
- Precision/Recall
- F_1-score

Can also use cross validation set to choose parameter ε



Anomaly detection vs. supervised learning

Anomaly detection vs. Supervised learning

Very small number (0 to 20) of positive examples $y = 1$; large number of negative examples ($y = 0$); Model $p(x)$ with just negative examples. Use positive examples for cv and test sets

Many different "types" of anomalies. Hard for any algorithm to learn (from just positive examples) what the anomalies look like.

Future anomalies may look nothing like any of the anomalous examples seen so far.

financial fraud

Large number of positive and negative examples.

≥ 20 positive examples

Enough positive examples for algorithm to get a sense of what positive examples are like.

Future positive examples likely to be similar to ones in training set.

email spam

Anomaly detection vs. Supervised learning

- Fraud detection

- Manufacturing:

Finding new previously unseen defects.
(e.g. aircraft engines)

- Monitoring machines in a data center
- Security applications

- Email spam classification

- Manufacturing:

Finding known, previously seen defects
(e.g. scratches on smartphones, $y = 1$)

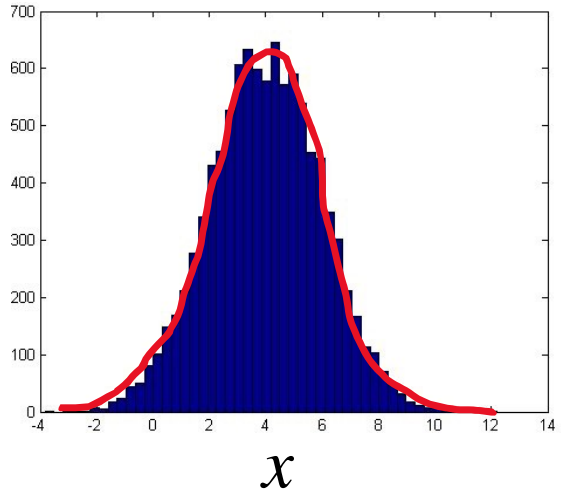
- Weather prediction (sunny/rainy/etc.)

- Diseases classification



Choosing what features to use

Non-gaussian features



$p(x_1; \mu_1, \sigma_1^2)$

`plt.hist(x)`

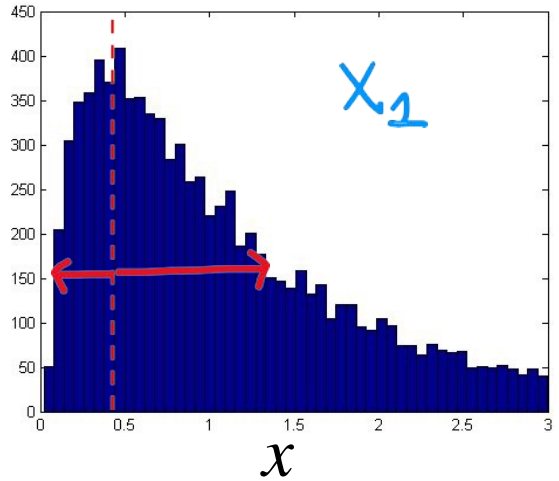
$x_1 \leftarrow \log(x_1)$

$x_2 \leftarrow \log(x_2 + 1)$ $\log(x_2 + c)$

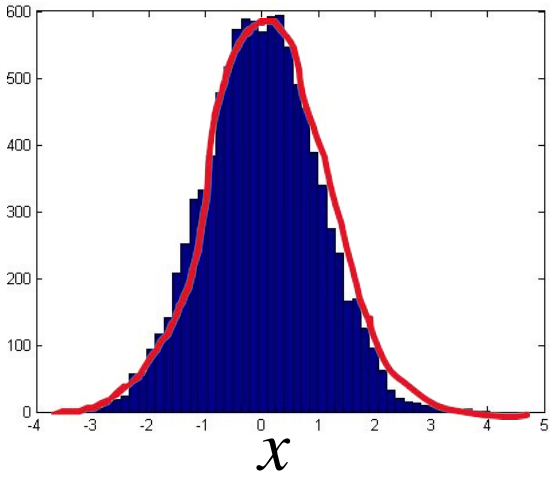
$x_3 \leftarrow \sqrt{x_3} = x_3^{1/3}$

$x_4 \leftarrow x_4^{3}$

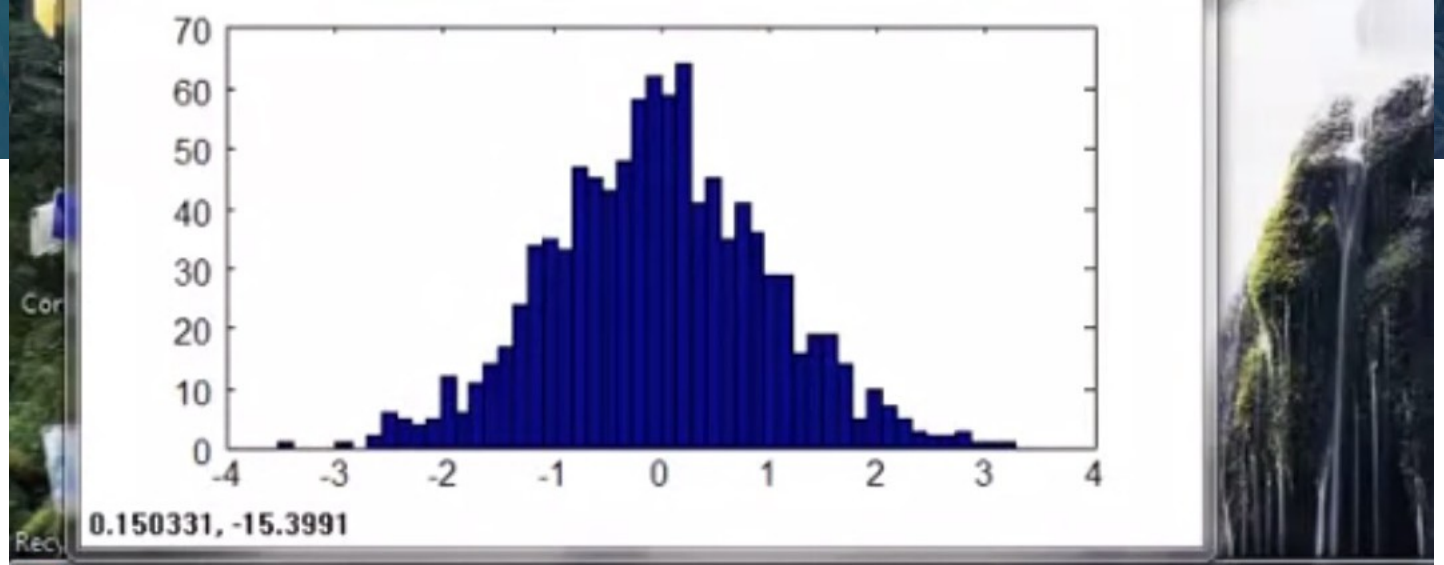




`np.log(x)`



Andrew Ng



Octave-3.2.4

```
   1000        1

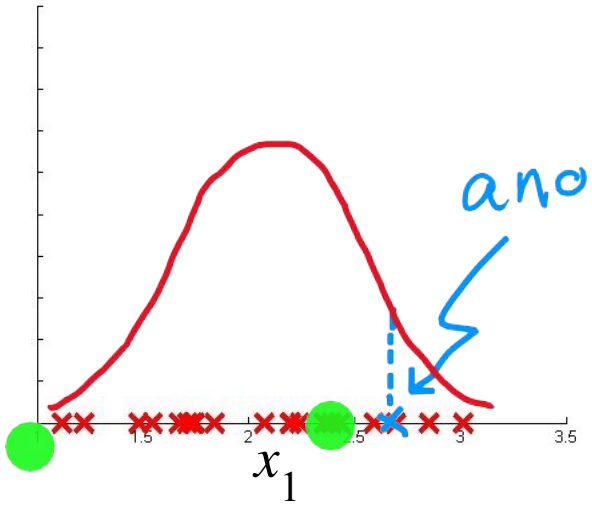
octave-3.2.4.exe:6> hist(x)
octave-3.2.4.exe:7> hist(x,50)
octave-3.2.4.exe:8> hist(x.^0.5, 50)
octave-3.2.4.exe:9> hist(x.^0.2, 50)
octave-3.2.4.exe:10> hist(x.^0.1, 50)
octave-3.2.4.exe:11> hist(x.^0.05, 50)
octave-3.2.4.exe:12> xNew = x.^0.05;
octave-3.2.4.exe:13> hist(log(x),50)
octave-3.2.4.exe:14> xNew = log(x);
```

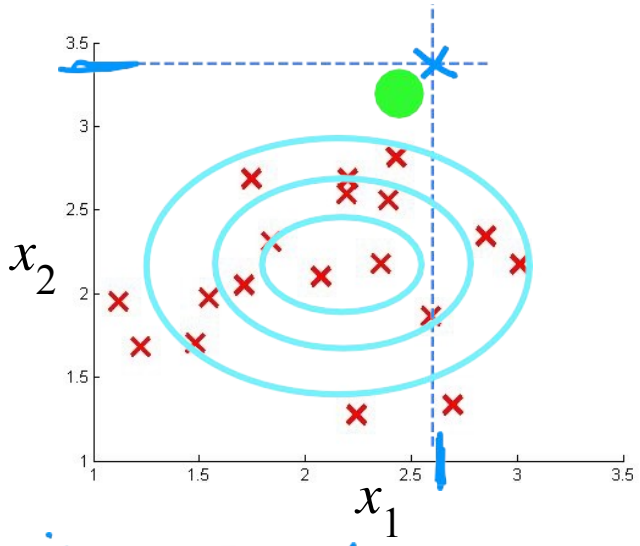
Error analysis for anomaly detection

Want $p(x) \geq \varepsilon$ large for normal examples x.
$p(x) < \varepsilon$ small for anomalous examples x.

Most common problem:

$p(x)$ is comparable for normal and anomalous examples.
($p(x)$ is large for both)





x_1 num transactions

x_2 typing speed

Monitoring computers in a data center

Choose features that might take on unusually large or small values in the event of an anomaly.

x_1 = memory use of computer

x_2 = number of disk accesses/sec

x_3 = CPU load

x_4 = ~~network traffic~~ CPU load / network traffic

high low

not unusual

$$x_1 = \frac{(\text{CPU load})^2}{\text{network traffic}}$$

Deciding feature choice based on $p(x)$

Large for normal examples;

Becomes small for anomaly in the cross validation set



Q & A